BIOMANUFACTURING PROCESS ANALYTICAL TECHNOLOGY

Programmable Growth and Oxygen-Transfer Monitoring with paceval

Reusable mathematical execution graphs for an illustrative fed-batch bioreactor, with independently checked model versions and transparent scientific boundaries.



Concept illustration; not a computed research result.

RESEARCH OUTCOME

Twenty-four distinct mathematical models and related analytical functions were executed through the original paceval MCP service (reported version 4.26). The evidence covers an 81-point sensor-state grid and 16 synthetic replay snapshots. All 515 retained ordinary numerical outputs passed an independent high-precision reference check.

PRECISE VERIFICATION BOUNDARY

Eighteen printed TINC interval records were audited. Fifteen showed direct inclusion under an exact-binary-input interpretation; three collapsed printed score intervals require investigation using typed local SDK bounds. The C++ source passes syntax checks but could not be linked or executed without a compatible original Linux library. No GMP qualification is claimed.

Scientific Research Demonstrator | 14 November 2026

1. Industrial use case and current state of the art

Modern biomanufacturing uses controlled bioreactors to cultivate microorganisms or mammalian cells for biologic products, enzymes and other valuable substances. Operators monitor substrates, viable or dry-cell biomass, pH, dissolved oxygen (DO), gases and feeding conditions. A process analytical technology (PAT) layer transforms these measurements into interpretable estimates, trends and alerts. Unlike a single sensor alarm, a mathematical monitoring layer can make explicit why a process is approaching a limitation.

The focal scenario is a hypothetical fed-batch reactor producing a biological product. A set of synthetic process snapshots supplies substrate S, dissolved oxygen DO, biomass X, aeration effectiveness kLa and liquid feed F. A mathematical model calculates specific growth, substrate consumption, oxygen uptake (OUR), oxygen transfer (OTR) and instantaneous mass-balance slopes. Numerical guard expressions combine the estimated oxygen transfer capacity with a direct DO concentration threshold. The purpose is reproducible function evaluation, not the automatic adjustment of a physical reactor.

What industry uses today

Established solutions combine process probes, off-gas analysis, Raman or near-infrared measurements, soft sensors, mechanistic or hybrid models, and manufacturing execution/quality systems. FDA PAT guidance (2004) encourages scientifically grounded process understanding. ICH Q13 (2022/2023) covers monitoring and control considerations for continuous manufacturing, including therapeutic proteins. Such references set a conceptual context and do not certify this software [1,2].

A 2024 study considered real-time oxygen-uptake monitoring in CHO antibody-production reactors. Reviews published in 2025 and 2026 discuss in-situ bioreactor instrumentation, oxygen sensing and soft-sensor integration, along with calibration drift and model transportability challenges [3-6]. This project isolates the reusable mathematical expressions within that wider pipeline; it does not attempt to replace spectroscopy, instrumentation or a validated plant historian.

Research question

Can the original paceval engine execute and reuse biologically meaningful kinetic, oxygen and fed-batch balance expressions, their versioned alternatives and explicit warning functions, while retaining an independently auditable numerical trail? The experiment answers that mathematical functional question for deterministic inputs. It does not assess actual product quality, cell viability or bioreactor dynamics against physical observations.

2. Defined kinetic and mass-balance model

The source WP 05 proposes Monod-type growth with oxygen limitation, substrate uptake, constant cell-specific oxygen demand and a gas-liquid oxygen balance. The current study retains those four ideas and adds a separately versioned substrate-inhibition/growth-coupled oxygen-uptake scenario, two process constraints, material-balance rates and analytical derivatives.

Version v1: saturating growth

mu1 = umax * S/(Ks + S) * DO/(Ko + DO)

rS1 = mu1 * X / Yxs OUR1 = qO2 * X

Version v2: hypothetical substrate-inhibition extension

mu2 = umax * S/(Ks + S + S^2/Ki) * DO/(Ko + DO)

OUR2 = (qO2 + qG*mu2) * X rS2 = mu2 * X / Yxs

These are illustrative parameterizations, not identified CHO kinetic laws. The v2 formulation has a potentially nonmonotonic substrate response. Its constants were chosen for an algebraic demonstration and must not be interpreted as organism-specific values.

Fed-batch instantaneous mass-balance rates

OTR = kLa * (DOstar - DO)

dX/dt = (mu - F/V) * X

dS/dt = (F/V)*(Sfeed - S) - mu*X/Yxs

dDO/dt = OTR - OUR - (F/V)*DO

V is the current reactor volume, assumed supplied per snapshot. No time integration is performed on the synthetic replay. Feed is assumed oxygen-free, there is no effluent, and kLa is an effective lumped coefficient. In a physical simulation V changes with feed, and gas, pH, metabolic shifts and volume constraints require additional equations.

Dimensions and domains

S and Sfeed are g substrate/L; X is g dry biomass/L; DO and DOstar are g oxygen/L. Growth rates are 1/h, F is L/h, V is L, kLa is 1/h, Yxs is g biomass/g substrate, qO2 is g oxygen/(g biomass·h), and qG is g oxygen/g biomass. Ks, Ko and Ki carry concentration units as appropriate. The required positive denominators and DOstar greater than DO are validated on the host before evaluating any graph.

3. Oxygen-limitation monitoring and constraints

The oxygen driving force (DOstar - DO) and transfer coefficient kLa produce a rate OTR. A rate-only oxygen margin is not a complete dissolved-oxygen safety rule: OTR may exceed OUR while the current DO is already below an operating floor.

Moxygen_v = OTR - OUR_v [g O2 / L / h]

MDO = DO - DOmin [g O2 / L]

Score_v = min(MDO, tau * Moxygen_v) [g O2 / L]

Guard_v = (DO >= DOmin) AND (Moxygen_v >= 0) AND (S >= 0)

A host-defined monitoring decision interprets the paceval guard. Equality at the boundary counts as mathematically satisfying the non-strict >= constraints. An adequate positive score only means the deliberately chosen mathematical conditions hold for the given inputs. It is not an independent industrial acceptance or product-release statement.

Analytical sensitivities and model comparisons

d(mu2)/dS = umax * DO/(Ko+DO) * (Ks-S^2/Ki)/(Ks+S+S^2/Ki)^2

d(dDO/dt)/d(kLa) = DOstar - DO

kLa_required_v2 = OUR2/(DOstar - DO)

Delta_mu = mu2 - mu1

The host is responsible for domain checks, input schema, units, ordering, graph-handle lifecycle and stateful procedures. The engine executes all explicitly registered formulas, including composite scoring, comparisons, sensitivity functions, minimum/maximum operators and logical guards. No automatic differentiation is claimed.

Why versioning matters

The experiment retains two independent persistent graph versions instead of mutating a single expression. Their differences are scientifically interpretable: v1 increases saturatingly with substrate; v2 can decline at higher substrate due to an explicitly added inhibition term. More flexible formula deployment does not make either model automatically correct for a production strain.

4. Actually executed paceval graph architecture

Figure 1 | Industrial PAT computation architecture (conceptual)

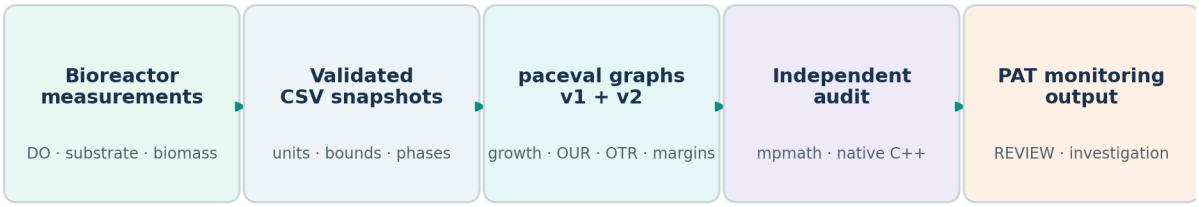


Figure 1. Conceptual architecture. This schematic is not an executed bioreactor measurement.

The live paceval MCP service reports version 4.26 in the recorded responses. Twenty-four independently registered mathematical graphs were evaluated together for 81 parameter-grid cases and 16 synthetic replay cases. Selected graphs were reused for a further 16-point verification call. Three interval-enabled graphs supplied six additional points each. A separate operator batch tested infix minimum/maximum, logical expressions, signed remainder and mathematical modulo.

Functional group	paceval graph responsibility	Host responsibility
Growth & consumption	v1/v2 mu, rS, dX, dS	Units, replay phase and data schema
Oxygen rates	OUR v1/v2, OTR, dDO	Validate DOstar > DO and finite values
Quality rules	Margins, 0/1 guards and min score	Define REVIEW/INVESTIGATE semantics
Model analysis	Sensitivity, delta and kLa requirement	Select and document model version
Numerical audit	TINC output in remote execution	High-precision checks and traceability

Local C++ integration

The supplied C++17 project includes the original paceval_main.h and paceval_main.cpp files without replacing the engine. It implements paceval_CreateComputation, scalar checks, per-graph batch checks, multi-graph batch evaluation, error checking, TINC output fields and resource cleanup. Numeric input and output are delegated to paceval_dConvertStringToFloat and paceval_dConvertFloatToString, after a strict decimal-literal grammar has rejected expression text in CSV fields. CMake requires a separately supplied ABI-compatible Linux runtime and does not redistribute a shared library.

This C++ code was syntax-checked with a standard Linux C++ compiler. It was not linked, executed or CTest-validated because the compatible original shared runtime was not available. The measured numerical results in this paper come from real remote MCP calls, never from an unexecuted local binary.

5. Deterministic research workload and execution evidence

All measurements are synthetically specified. The 16 replay rows represent a sequence of candidate sensor snapshots across distinct hypothetical feeding and oxygen-transfer phases; they are NOT a dynamically integrated biological trajectory. The 81-state grid varies biomass and dissolved-oxygen concentration while holding the other inputs fixed. Both data sets use explicit concentration and flow units and retain the original input text separately from the generated numerical output.

Executed scope	Verified record
Original v1/v2 and supporting graph registrations	24 registered; 24 executed
Parameter grid	81 cases × 24 graphs = 1,944 graph-point evaluations
Synthetic replay	16 cases × 24 graphs = 384 graph-point evaluations
Additional replay selections	6 reused graphs × 16 cases = 96 evaluations
Interval-enabled sub-study	3 additional graphs × 6 cases = 18 interval outputs
Ordinary independent checks	515 retained recorded values; all passed
Printed TINC interval checks	15 inclusive; 3 unresolved from printed endpoints
Local SDK/CTest	Source syntax checked; runtime execution unavailable

Interpretation of a representative snapshot

At replay hour 10, the synthetic inputs contain DO = 0.0016 g/L and biomass X = 6.5 g dry mass/L. paceval evaluates v2 oxygen uptake as approximately 0.30129 g oxygen/L/h and oxygen transfer as 0.352 g oxygen/L/h. The resulting positive oxygen margin does not eliminate the need for a separate concentration floor. At hours 11 and 12, the supplied DO values fall below the illustrative DOmin of 0.0015 g/L even though oxygen transfer remains adequate. At hour 13, equality at DOmin yields a zero composite score and a true non-strict logical guard.

All rates and decision thresholds are demonstrator inputs. The synthetic time index must not be mistaken for kinetic integration, predictive process behavior, process capability, or a validated QbD design space.

6. Growth and oxygen-transfer results

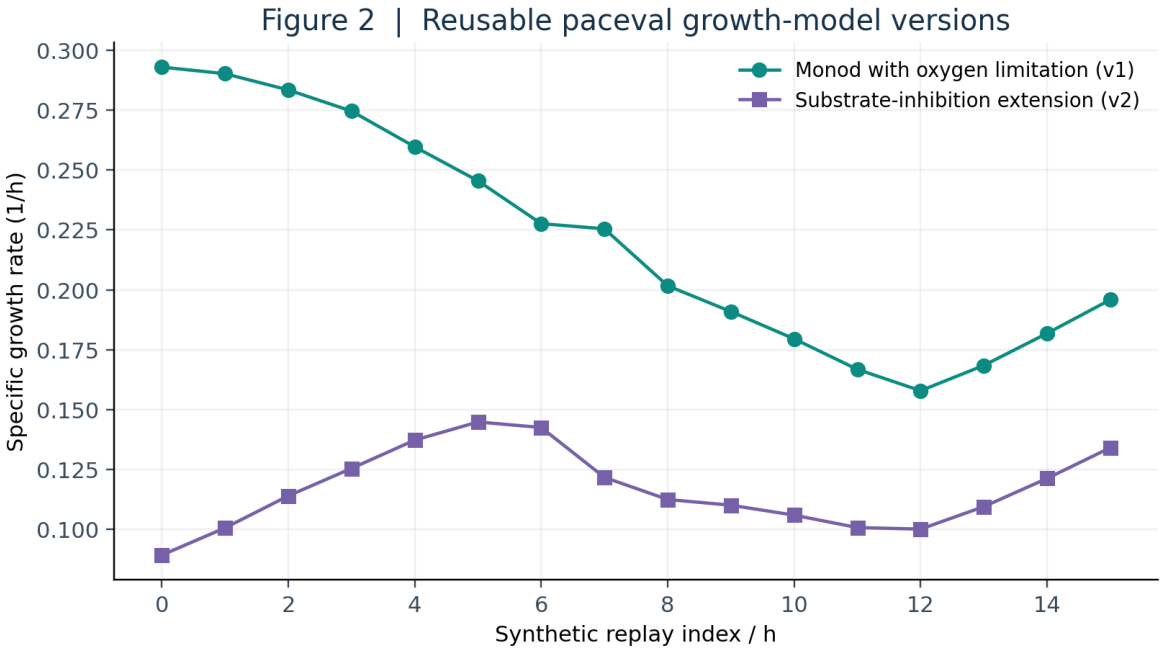


Figure 2. paceval MCP v4.26 growth-model values. The distinct v1/v2 responses show the functional effect of the model version, not experimental fit quality.

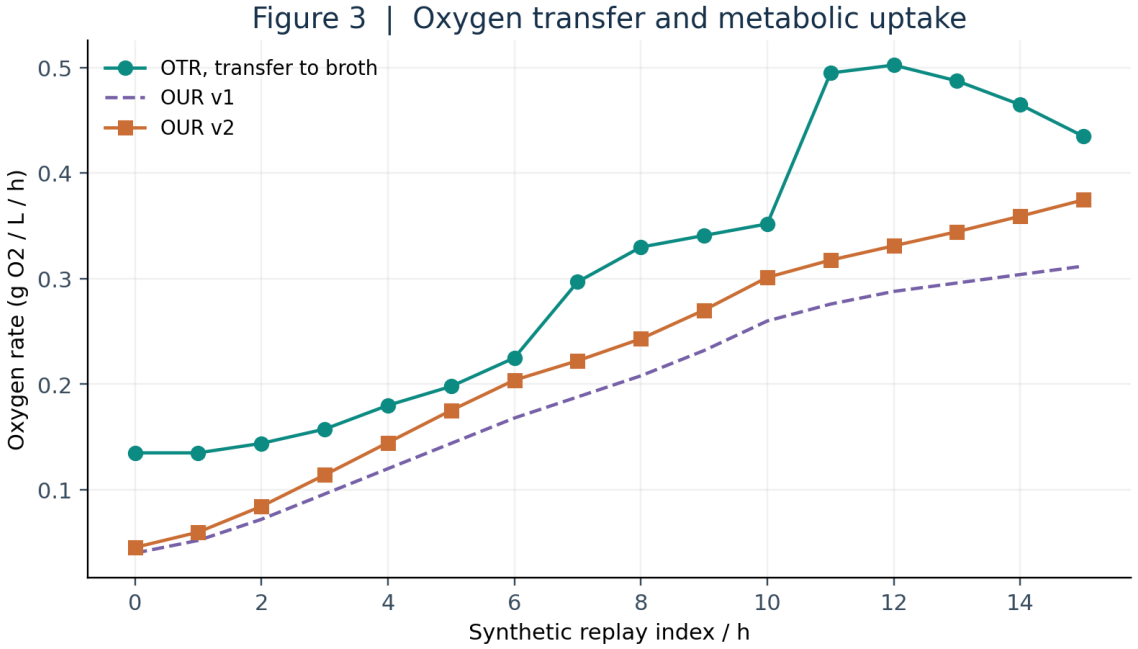


Figure 3. OTR and two OUR versions returned by paceval for the synthetic snapshot series.

7. Process margins and monitoring decisions

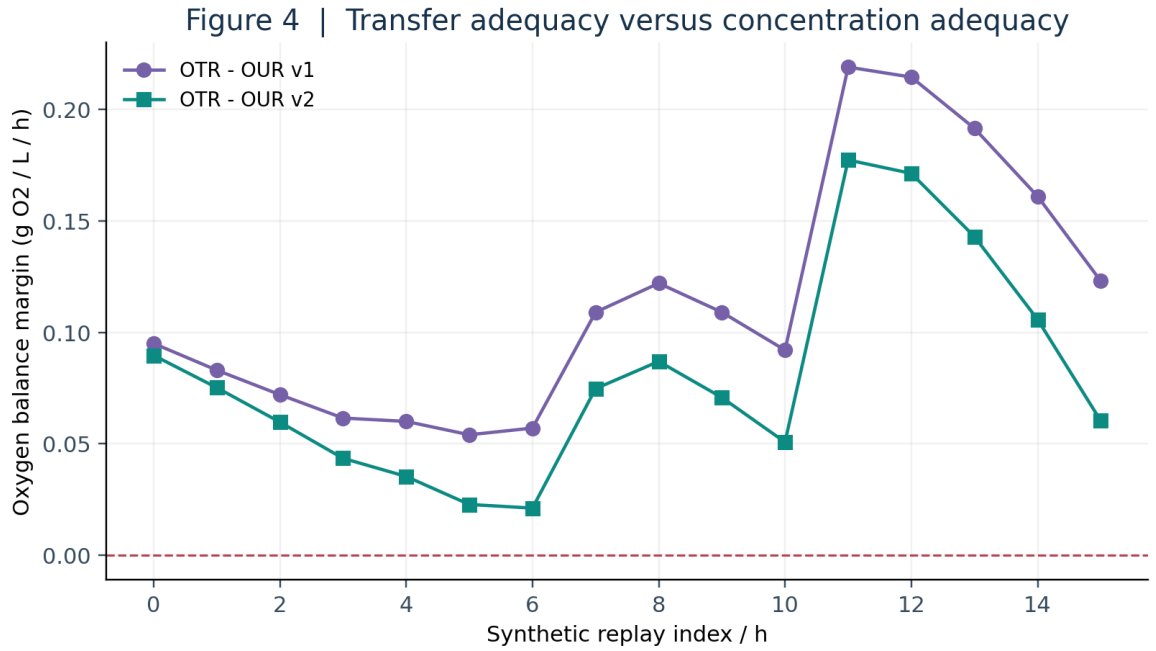


Figure 4. Independent paceval margins quantify the difference between gas-liquid oxygen transfer and cellular uptake.

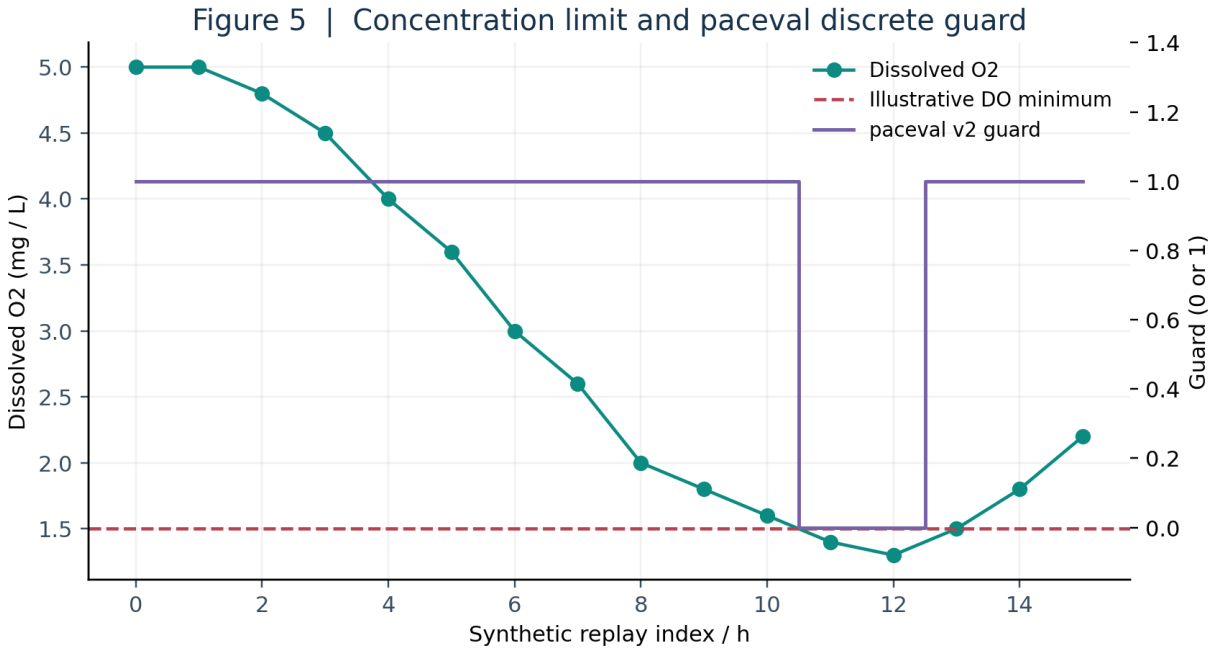


Figure 5. Synthetic DO input with the v2 paceval-computed logical guard. The guard becomes false when the DO input is below the selected illustrative floor.

8. Multidimensional process-state analysis

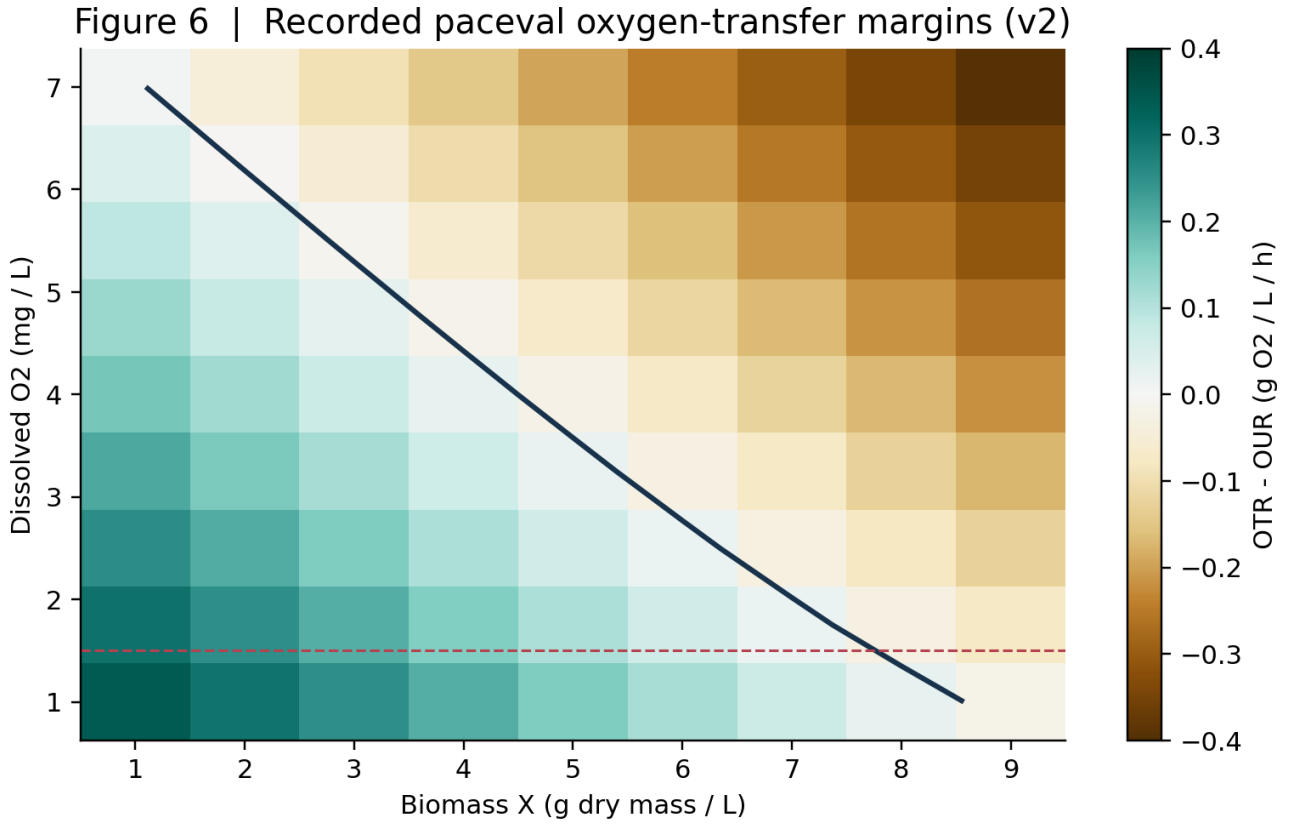


Figure 6. The 81 actually executed paceval points are displayed as a parameter map with the zero-margin contour. The dataset is synthetic, and interpolation between samples is illustrative only.

The parameter grid demonstrates that more biomass demands greater oxygen-transfer capacity, while changes in dissolved oxygen modify both mass-transfer driving force and growth-coupled oxygen uptake. The zero-margin line is a mathematical limit of the stipulated reduced-order model, not a production-approved process boundary.

9. Composite score surface

Figure 7 | paceval composite score surface (v2)

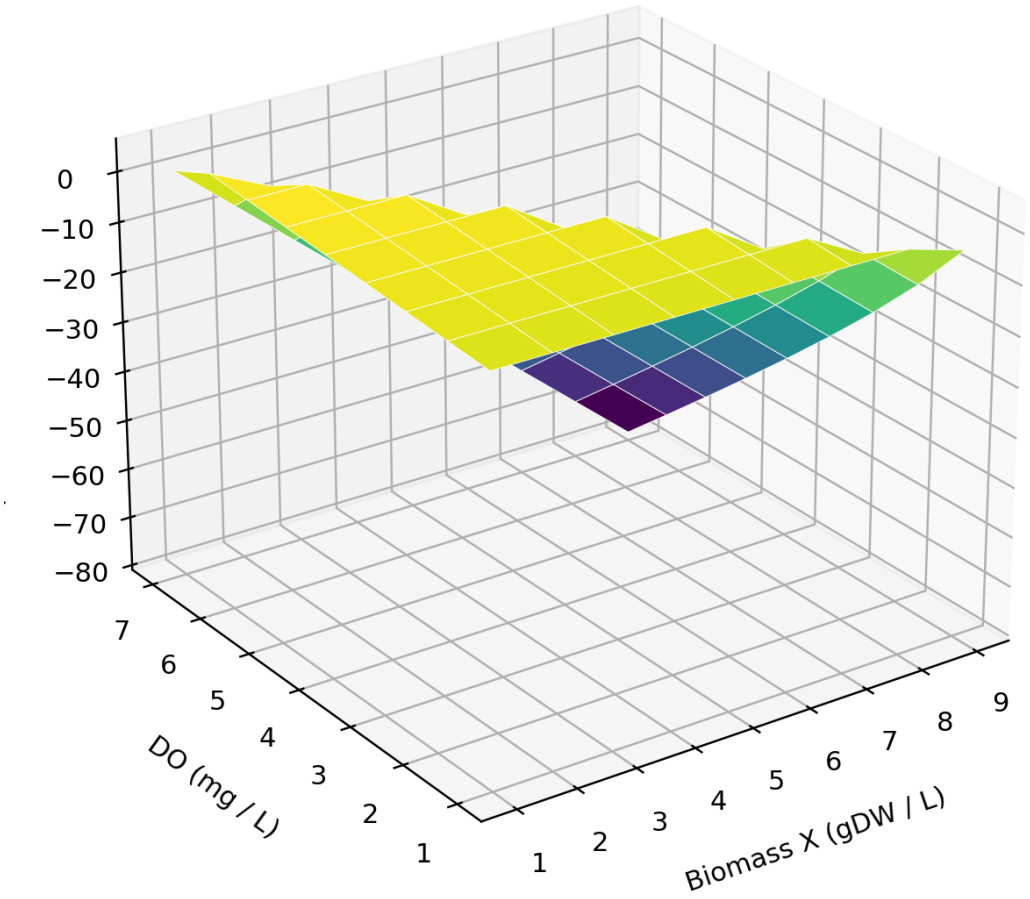


Figure 7. Recorded paceval v2 composite scores on the same 81 state combinations. Positive and negative surface values encode the chosen mathematical constraints after scaling the oxygen-rate margin by the illustrative horizon tau.

The min operator is executed inside paceval. The surface exposes non-smooth transitions between DO-concentration limitation and transfer-rate limitation. The transition is a model and policy feature, not evidence of a discontinuous cell-physiology response.

10. Required oxygen transfer and sensitivity

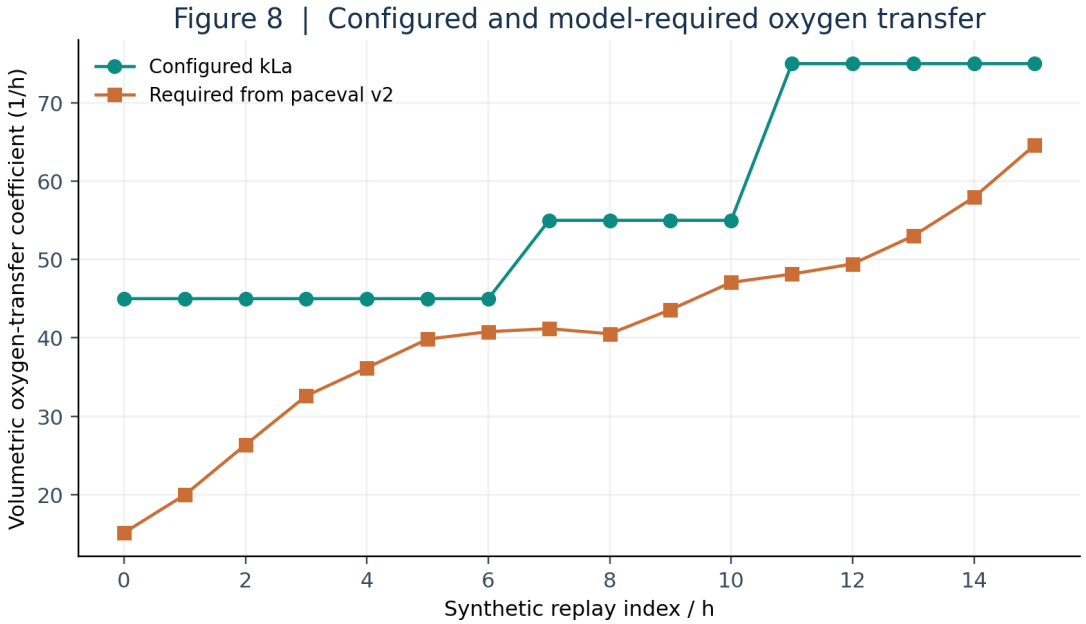


Figure 8. Real paceval evaluations of the kLa requirement under v2 and the supplied phase-specific kLa input.

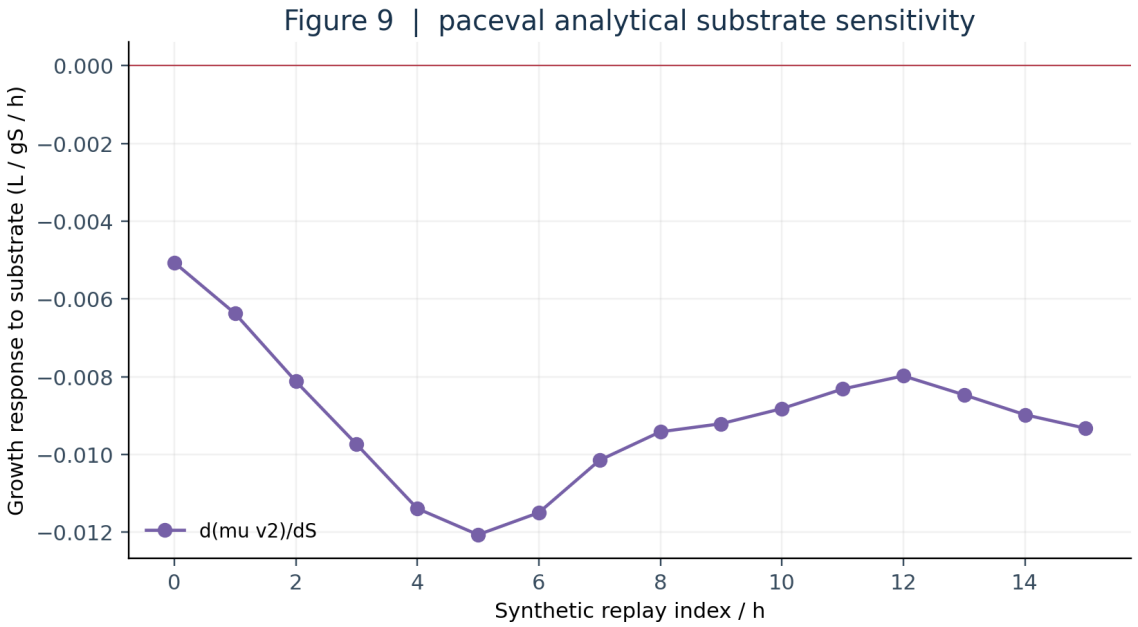


Figure 9. Explicit analytical paceval expression for substrate sensitivity in v2. The negative values in these synthetic snapshots reflect substrate-inhibition behavior.

11. Trusted interval output and independent verification

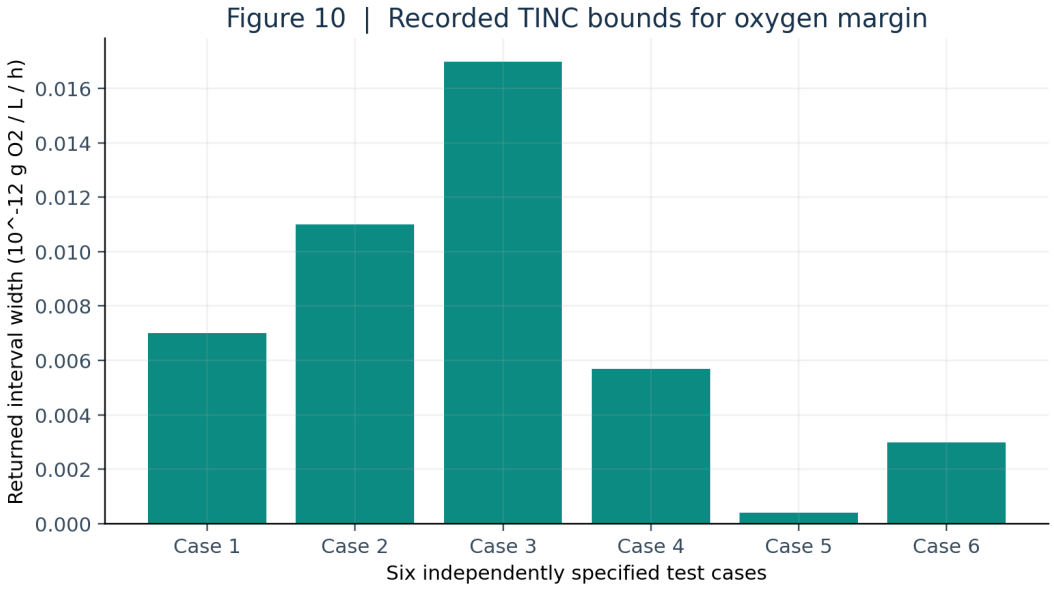


Figure 10. Widths of reported TINC interval bounds for six oxygen-margin checks. These are transmitted numerical bounds from remote paceval, not a claim of global parameter-interval certification.

A separate independent 90-digit mpmath implementation uses the exact binary64 values of all incoming numerical parameters, not the same paceval formula strings. All 515 ordinary retained numerical outputs in the replay and grid audit passed their predefined comparison rule. For TINC bounds, 15 of the 18 reported decimal intervals directly enclosed the corresponding independently computed value. Three v2 score cases were returned as identical printed decimal lower and upper endpoints at the selected threshold. The exact-binary-input reference lies outside these literal decimal endpoints. They remain unresolved until typed double bounds from the local SDK are retrieved.

This finding is not disguised as full TINC certification or evidence that the engine’s underlying binary interval implementation is invalid. The available remote textual output is insufficient to distinguish an interval-computation defect from a serialization/rounding limitation. Local ABI-compatible execution, raw binary interval archives, and directed-rounding regression tests are a necessary follow-up.

12. Scientific limitations, reproducibility and references

Established facts versus hypotheses

What is executed: reusable paceval graph evaluation, v1/v2 kinetic expressions, oxygen rates, fed-batch instantaneous derivative expressions, composite scoring, logical 0/1 conditions, a parameter grid, a synthetic replay and independently checked recorded results. What is not established: accurate behavior for a specific mammalian cell line, sensor-calibrated parameter estimates, validated oxygen-transfer identification, closed-loop controller stability, GMP compliance, process capability or a release-by-exception decision.

The baseline is a reduced-order mathematical model. Oxygen concentrations, growth rate, feed composition, respiration, process volume and mass transfer interact with hidden states, adaptation, pH, by-products and gas-liquid dynamics in real vessels. Quantitative scaling requires industrial sensor validation, identifiability analysis, uncertainty models, dynamic integration and domain-appropriate acceptance criteria. FDA and ICH guidance should be interpreted within the full applicable quality system; this research is not a regulatory submission.

Reproduction

Run **python validation/high_precision_reference.py** to regenerate the independent audit. Run **python scripts/make_figures.py** to regenerate scientific graphics from the recorded remote paceval files. To execute the original local SDK route, provide an ABI-compatible x86-64 Linux library and run **bash scripts/build_and_test.sh /absolute/path/libpaceval_linux_sharedLIB.so**. The latter remains an unexecuted test path in the supplied release. API signatures should be checked against the included original SDK files.

Selected external references

[1] FDA, PAT: A Framework for Innovative Pharmaceutical Development, Manufacturing, and Quality Assurance, 2004. FDA guidance.

[2] FDA / ICH Q13, Continuous Manufacturing of Drug Substances and Drug Products, 2022-2023. Q13 guidance.

[3] A real-time oxygen uptake rate monitoring approach suitable for the antibody production process, 2024. Journal link.

[4] Benchmarking and Trends Around In-situ and Ex-situ Sensors in Bioreactors, 2025. DOI.

[5] Optical Fiber pH and Dissolved Oxygen Sensors for Bioreactor Monitoring: A Review, published 2025 (2026 issue). DOI.

[6] Less Is More: Practical Insights into Multivariate Regression Models for Raman Spectroscopy in Bioprocess Monitoring, 2025. PubMed.

[7] paceval Mathematical Engine, Original API reference (4.25 documented). Live MCP service reported v4.26 for this study.